

Impulse Invariance Method:

The system transfer function of analog filter, $H_a(s)$ is given:-

$$H_a(s) = \frac{A_1}{s-p_1} + \frac{A_2}{s-p_2} + \frac{A_3}{s-p_3} + \dots$$

$$\therefore H_a(s) = \sum_{i=1}^N \frac{A_i^*}{s-p_i}$$

where, $A_i^* = A_1, A_2, A_3, \dots, A_N$ are coeff. of PFE.

$p_i^* = p_1, p_2, p_3, \dots, p_N$ are the poles

$s =$ Laplace operator.

The Inverse Laplace Transform is given by

$$h_a(t) = \sum_{i=1}^N A_i^* e^{p_i^* t}$$

put $t = nT$. Then

$$h_a(nT) = h(n) = \sum_{i=1}^N A_i^* e^{p_i^* nT}$$

$$\therefore h(n) = \sum_{i=1}^N A_i^* e^{p_i^* nT}$$

where, $T =$ sampling time.

Taking z-transform of $h(n)$, we get;

$$H(z) = \sum_{n=0}^{\infty} h(n) z^{-n}$$

$$\Rightarrow H(z) = \sum_{n=0}^{\infty} \left(\sum_{i=1}^N A_i e^{p_i n T} \right) z^{-n}$$

$$\Rightarrow H(z) = \sum_{i=1}^N A_i \sum_{n=0}^{\infty} e^{p_i n T} z^{-n}$$

$$\Rightarrow H(z) = \sum_{i=1}^N A_i \left(\sum_{n=0}^{\infty} (e^{p_i T} z^{-1})^n \right)$$

$$\Rightarrow H(z) = \sum_{i=1}^N A_i \times \frac{1}{1 - e^{p_i T} z^{-1}}$$

$$\therefore H(z) = \sum_{i=1}^N \frac{A_i}{1 - e^{p_i T} z^{-1}}$$

This is the required transfer function of digital filter.

(*) Relationship between s-plane & z-plane:

→ We have,

$$z = e^{P_i T} \quad \text{--- (1)}$$

where, $P_i = P_1, P_2, P_3, \dots, P_N$ are poles in s-plane.

$T = \text{Sampling Time.}$

Q1. Since $s = P_i$ so,

$$z = e^{sT} \quad \text{--- (2)}$$

Also, we have, $s = \sigma + j\Omega$

where, $\sigma = \text{attenuation factor}$

$\Omega = \text{analog frequency}$

$$\therefore z = e^{(\sigma + j\Omega)T} \quad \text{--- (3)}$$

we know that,

$$z = re^{j\omega} \quad \text{--- (4) in polar form,}$$

where, $r = \text{magnitude}$

$\omega = \text{frequency,}$

from (3) & (4), we get,

$$re^{j\omega} = e^{(\sigma + j\Omega)T}$$

$$\Rightarrow re^{j\omega} = e^{\sigma T} \cdot e^{j\Omega T}$$

On Comparing, We get;

$$r = e^{\sigma T}$$

$$\& e^{j\omega} = e^{j\Omega T} \quad \text{i.e. } \omega = \Omega T$$

(i) Case I: When $\sigma < 0$.

$$r = e^{\sigma T}$$

$$r = e^{-T} = \frac{1}{e^T}$$

$$\sigma < 0 \Rightarrow 0 < r < 1$$

$$\therefore \text{ } r \text{ is between } 0 \text{ to } 1$$

$$\text{i.e. } 0 < r < 1$$

Thus, $0 < r < 1$ indicates the interior part of unit unit circle. Therefore, L.H.S. of s-plane is mapped inside the unit circle.

(ii) Case II: When $\sigma = 0$

$$\therefore r = e^{0T} = 1$$

$$\text{i.e. } \sigma = 0 \Rightarrow r = 1$$

$\therefore \sigma = 0$ indicates $j\Omega$ axis and $r = 1$ indicates unit circle. Thus, $j\Omega$ axis in s-plane is mapped on the unit circle.

(iii) Case III: When $\sigma > 0$,

$$r = e^{\sigma T} = e^T$$

$$\therefore \sigma > 0 \Rightarrow r > 1$$

i.e. $\sigma > 0$ indicates R.H.S of s-plane and $r > 1$ indicates exterior part of unit circle. Thus, R.H.S of s-plane is mapped outside the unit circle

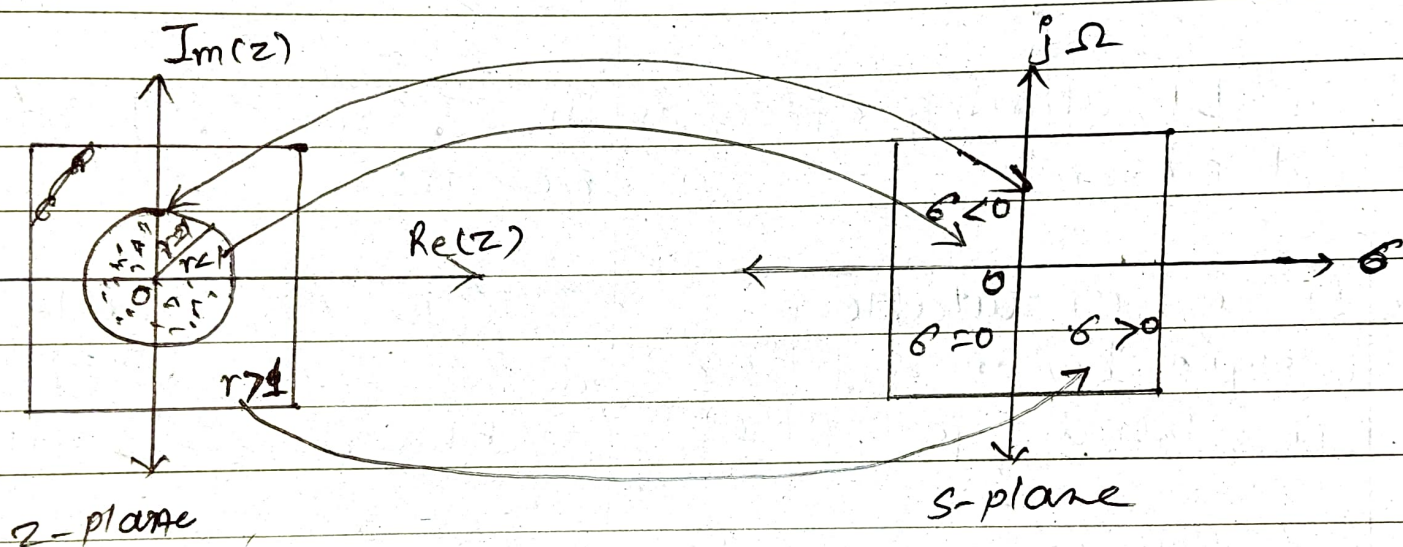


Fig. Illustration of the mapping $z = e^{sT}$

BLTM vs IIM

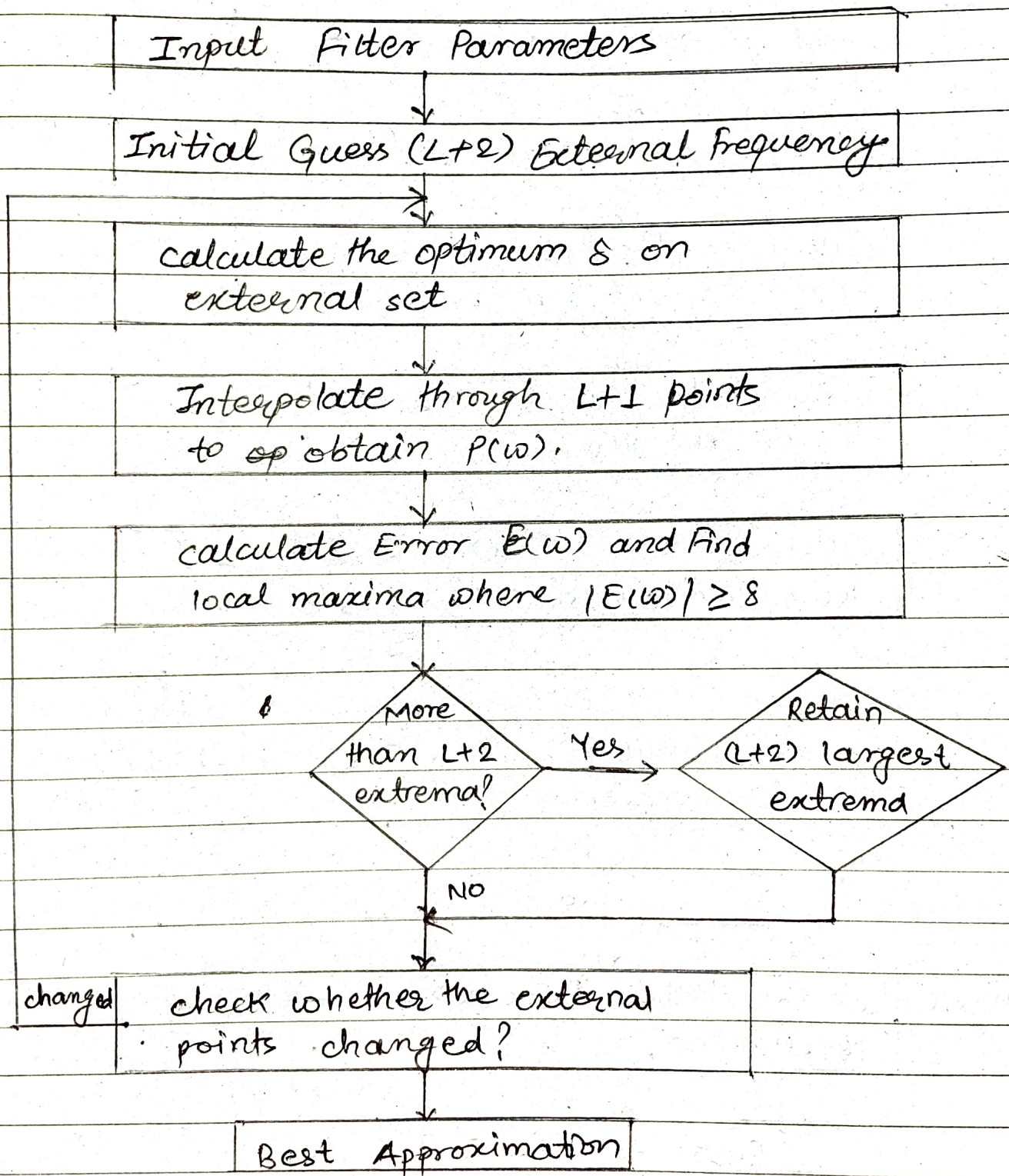
② Bilinear Transformation vs Impulse Invariance Method

S.N.	Bilinear Transformation	Impulse Invariance Method
(i)	In BLT, poles are transferred by using the expression: $s = \frac{2}{T} \left(\frac{z-1}{z+1} \right)$	In IIM, poles are transferred by using the expression: $\frac{1}{s - p_i} \rightarrow \frac{1}{1 - e^{p_i T} z^{-1}}$
(ii)	In BLT, Mapping is one-to-one.	In IIM, Mapping is many-to-one.
(iii)	In BLT, aliasing effect is not present.	In IIM, aliasing effect is present.
(iv)	BLT is suitable to design high pass filter and band reject filter.	IIM is not suitable to design high pass filter and band reject filter.
(v)	In BLT, poles as well as zeros can be mapped.	In IIM, only poles of the system can be mapped.
(vi)	In BLT, frequency warping effect is present.	In IIM, frequency warping effect is not present.
(vii)	In BLT, $\Omega = \frac{2}{T} \tan \frac{\omega}{2}$.	In IIM, $\Omega = \frac{\omega}{T}$.

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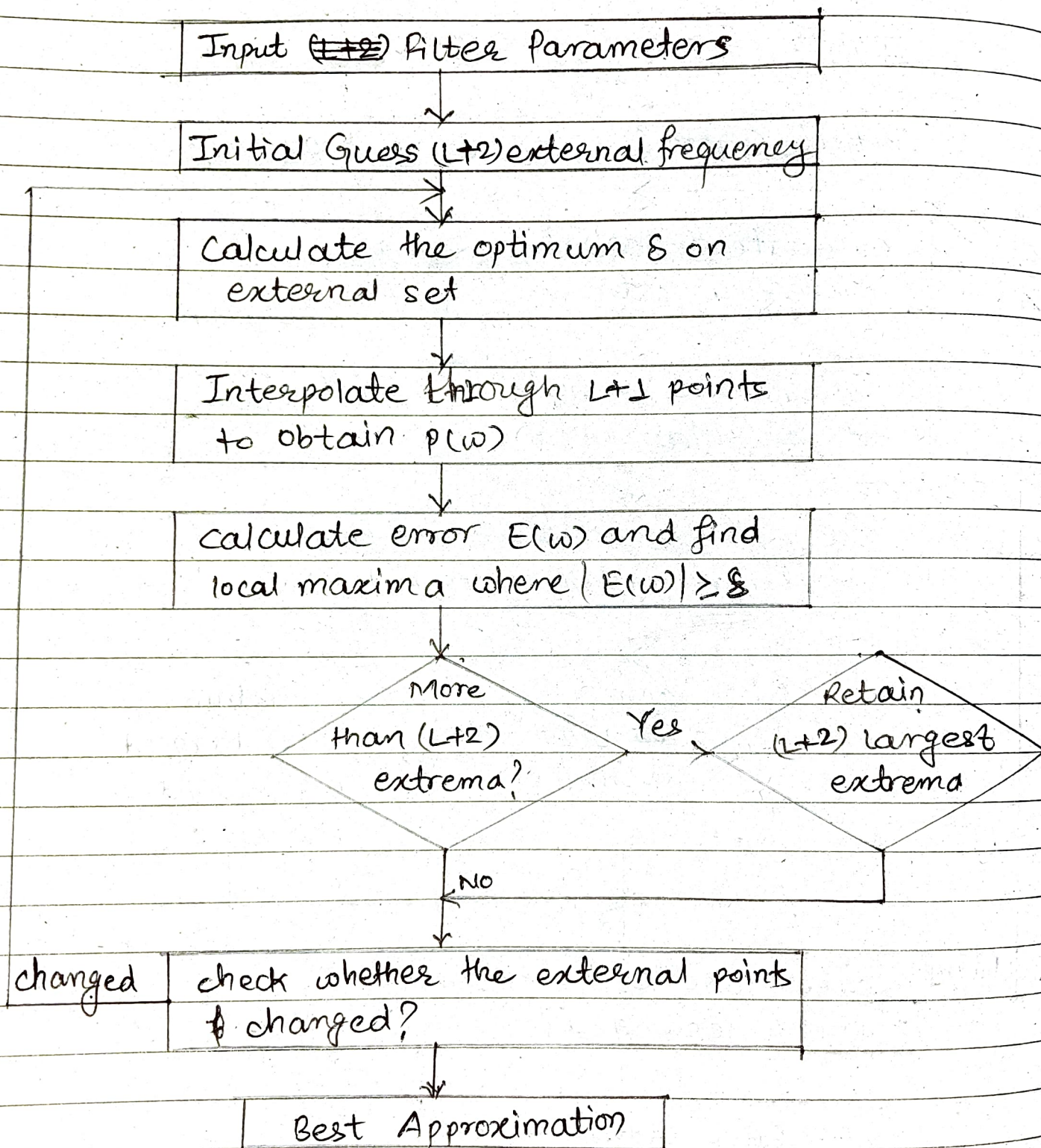
① Remez Exchange Algorithm:

→ Remez exchange algorithm is used to determine the locations of extremal frequencies.



(*) Remez Exchange Algorithm:

- 1 Remez exchange algorithm is used to determine the
1 locations of external frequencies.



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IIR vs FIR:

S.N.	IIR Filter	FIR Filter.
(i)	IIR stands for Infinite Impulse Response.	FIR stands for Finite Impulse Response.
(ii)	IIR has infinite impulse response duration.	FIR has finite impulse response duration.
(iii)	IIR uses feedback Mechanism.	FIR does not use feedback mechanism.
(iv)	IIR may be stable based on pole position.	FIR is always stable.
(v)	IIR is generally designed for non-linear phase.	FIR can be designed for linear phase.
(vi)	IIR requires less memory.	FIR requires more memory.
(vii)	IIR requires less computation due to fewer coefficients.	FIR requires more computation due to more coefficients.
(viii)	IIR Filter design methods: Impulse Invariance method, Bilinear Transformation methods etc.	FIR Filter design methods: window method, frequency sampling, Remez exchange algorithm etc.

Chapter-4

Multiplication of two DFT $\Leftarrow$ ~~Convolution of Circular Convolution~~
Let us consider two discrete time sequence $x_1(n)$ & $x_2(n)$.
Then the DFT of $x_1(n)$ & $x_2(n)$ are as follows:

$$X_1(K) = \sum_{n=0}^{N-1} x_1(n) \cdot e^{-j2\pi Kn/N} \quad (1)$$

where, $K = 0, 1, 2, \dots, N-1$

$$X_2(K) = \sum_{n=0}^{N-1} x_2(n) \cdot e^{-j2\pi Kn/N} \quad (2)$$

where $K = 0, 1, 2, \dots, N-1$

from (1) & (2), we get:-

$$Y(K) = X_1(K) \cdot X_2(K) \quad (3)$$

$$\begin{aligned} \therefore Y(K) &= \left(\sum_{n=0}^{N-1} x_1(n) e^{-j2\pi Kn/N} \right) \cdot \left(\sum_{n=0}^{N-1} x_2(n) e^{-j2\pi Kn/N} \right) \\ &= \sum_{n=0}^{N-1} \end{aligned}$$

Let the IDFT of $Y(K)$ be $y(m)$. Then,

$$y(m) = \frac{1}{N} \sum_{K=0}^{N-1} Y(K) e^{j2\pi Km/N} \quad (4)$$

from

$$\Rightarrow y(m) = \frac{1}{N} \sum_{k=0}^{N-1} x_1(k) \cdot x_2(k) e^{j \frac{2\pi k m}{N}} \quad (\text{from (3)})$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{n=0}^{N-1} x_1(n) e^{-j \frac{2\pi k n}{N}} \right] \left[\sum_{l=0}^{N-1} x_2(l) e^{-j \frac{2\pi k l}{N}} \right] e^{j \frac{2\pi k m}{N}}$$

$$y(m) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{k=0}^{N-1} x_2(k) \left[\sum_{l=0}^{N-1} e^{j \frac{2\pi (m-n-l)k}{N}} \right] \quad (5)$$

let us consider the last term of equation,

$$e^{j \frac{2\pi (m-n-l)k}{N}} = \left[e^{j \frac{2\pi (m-n-l)}{N}} \right]^k$$

Since $N-1$

$$\sum_{k=0}^{N-1} a^k = \begin{cases} a^N & ; a = 1 \\ \frac{1-a^N}{1-a} & ; a \neq 1 \end{cases}$$

Thus,

$$\sum_{k=0}^{N-1} \left[e^{j \frac{2\pi (m-n-l)}{N}} \right]^k = \begin{cases} N & ; e^{j \frac{2\pi (m-n-l)}{N}} = 1 \\ \frac{1 - e^{j \frac{2\pi (m-n-l)}}{1 - e^{j \frac{2\pi (m-n-l)}{N}}}} & ; e^{j \frac{2\pi (m-n-l)}{N}} \neq 1 \end{cases}$$

~~we take~~, since m, l , and n are integers, so we take $e^{j2\pi(m-n-l)/N} = 1$.

$$y(m) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \cdot \sum_{l=0}^{N-1} x_2(l) \cdot N \quad \text{--- (6)}$$

~~we~~

If p is an integer, which can be positive or negative. Then,

$$m - \cancel{n} - l = -pN$$

$$\Rightarrow m - n + pN = l$$

$$\therefore l = m - n + pN$$

from (6),

$$y(m) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \cdot \sum_{l=0}^{N-1} x_2(m-n+pN) \cdot N$$

$$= \sum_{n=0}^{N-1} x_1(n) \cdot \sum_{l=0}^{N-1} x_2(m-n+pN)$$

$$= \sum_{n=0}^{N-1} x_1(n) \cdot \sum_{l=0}^{N-1} x_2((m-n)) \cdot N$$

$$\therefore y(m) = \sum_{n=0}^{N-1} x_1(n) \cdot x_2((m-n)) \cdot N, \quad m = 0, 1, \dots, N-1$$

Let us consider two discrete time sequences $x_1(n)$ & $x_2(n)$.
Then the DFT of $x_1(n)$ & $x_2(n)$ are as follows:

$$X_1(K) = \sum_{n=0}^{N-1} x_1(n) \cdot e^{-j2\pi Kn/N} \quad \text{--- (1)}$$

$$\therefore n = 0, 1, \dots, N-1$$

$$X_2(K) = \sum_{l=0}^{N-1} x_2(l) \cdot e^{-j2\pi Kl/N} \quad \text{--- (2)}$$

$$\therefore l = 0, 1, \dots, N-1$$

~~Let $y(m)$ be the~~
from (1) & (2), we get:

$$Y(K) = X_1(K) \cdot X_2(K) \quad \text{--- (3)}$$

Let $y(m)$ be the IDFT of (3) i.e. $Y(K)$. Then,

$$y(m) = \frac{1}{N} \sum_{K=0}^{N-1} Y(K) e^{j2\pi Km/N}$$

$$= \frac{1}{N} \sum_{K=0}^{N-1} X_1(K) \cdot X_2(K) \cdot e^{j2\pi Km/N}$$

$$= \frac{1}{N} \sum_{K=0}^{N-1} \sum_{n=0}^{N-1} x_1(n) e^{-j2\pi Kn/N} \cdot \sum_{l=0}^{N-1} x_2(l) e^{-j2\pi Kl/N} \cdot e^{j2\pi Km/N}$$

$$y(m) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l) \sum_{k=0}^{N-1} e^{j2\pi(m-n-l)k/N} \quad \text{--- (4)}$$

Let us consider the last term of the equation.

$$e^{j2\pi(m-n-l)k/N} = [e^{j2\pi(m-n-l)/N}]^k$$

We know,

$$\sum_{k=0}^{N-1} a^k = \begin{cases} a^N & ; a = 1 \\ \frac{1-a^N}{1-a} & ; a \neq 1 \end{cases}$$

For $a = 1$:

$$\sum_{k=0}^{N-1} a^k = N$$

$$\text{i.e. } \sum_{k=0}^{N-1} [e^{j2\pi(m-n-l)/N}]^k = N$$

where, $m, n, \& l$ are integers.

From (4),

$$y(m) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l) \quad N$$

$$\Rightarrow y(m) = \sum_{n=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l) \quad \text{--- (5)}$$

Since p is an integer, which may be positive or negative, so,

$$m - n - l = -pN$$

$$\Rightarrow -l = -pN + n - m$$

$$\therefore l = (m - n + pN)$$

From (5),

$$y(m) = \sum_{n=0}^{N-1} x_1(n) \cdot x_2(m-n+pN)$$

$$= \sum_{n=0}^{N-1} x_1(n) \cdot x_2((m-n))_N$$

$$\therefore y(m) = \sum_{n=0}^{N-1} x_1(n) \cdot x_2((m-n))_N$$

proved

This is the required

chapter-1

8 & # basic # Linear vs Circular Convolution:

S.N.	Linear Convolution	Circular Convolution
(i)	Linear Convolution is the multiplication of two sequences in time domain.	Circular Convolution is the multiplication of two sequences in frequency domain.
(ii)	The output length of linear convolution is $L+M-1$.	The output length of ^{circular} linear convolution is $L \oplus M$.
(iii)	In linear convolution, zero padding is not required.	In circular convolution, zero padding is required.
(iv)	Linear Convolution is used in LTI systems for real-world applications.	Circular Convolution is used in FFT-based filtering.
(v)	The linear convolution is given by $y(n) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(n-k)$	The circular convolution is given by $y(n) = \sum_{k=0}^{N-1} x(k) h((n-k))_N$

chapter-1

DSP and its basic elements:

→ Digital signal processing is the process of processing discrete signals that are sampled and represented as numbers.

→ The basic elements of DSP are as follows:

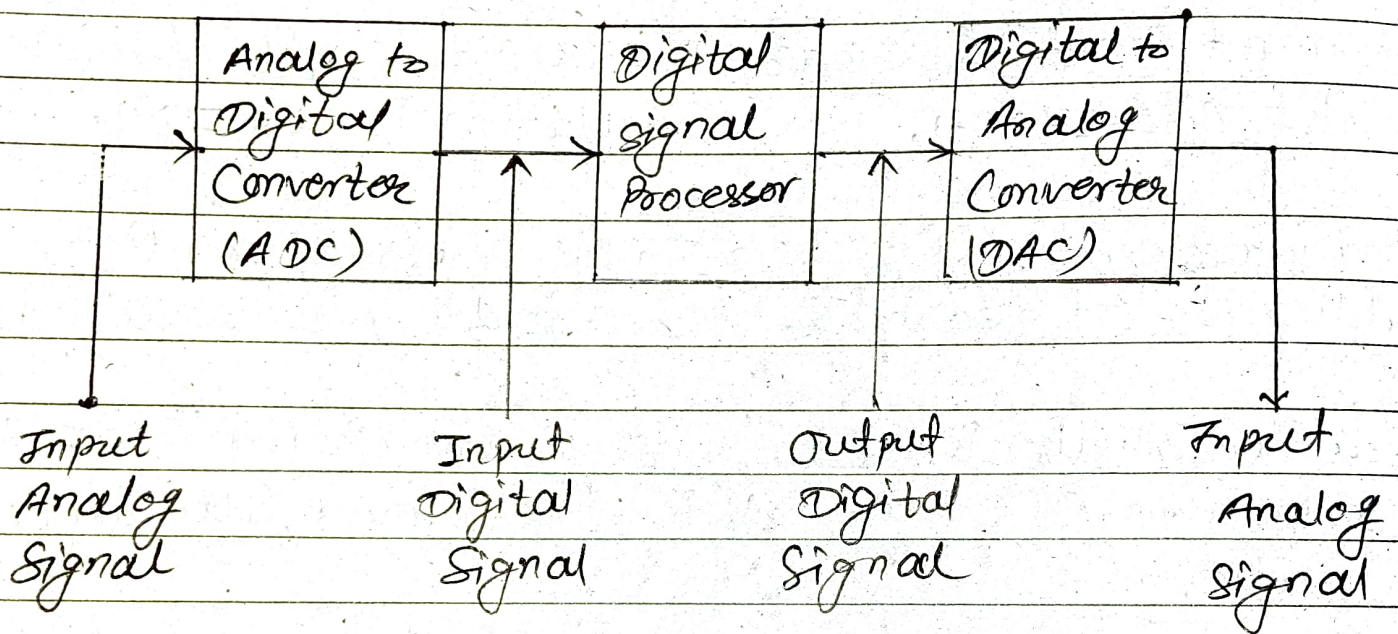


Fig. Basic elements of DSP

(i) Analog to Digital Converter (ADC):

⇒ Initially, the input analog signal passes through the analog to digital converter (ADC) to produce the input digital signal which is appropriate for digital signal processor.

(ii) Digital Signal Processor :

→ After that, the input of digital signal processor passes through digital signal processor to produce output digital signal.

(iii) Digital to Analog Converter: (DAC):

→ Some of the processed signals are required to back in their original analog form. So the output of the digital signal processor or output analog signal passes through the digital to analog converter (DAC) to produce output analog signal.

II Energy Signal vs Power Signal:

S. No.	Energy Signal	Power Signal
(i)	A signal is said to be energy signal if it has finite total energy and zero average power. i.e. $E < \infty$ $P = 0$	A signal is said to be power signal if it has finite average power and infinite total energy. i.e. $P > 0$ $E = \infty$
(ii)	The total energy is obtained by $E = \int_{-\infty}^{\infty} x(t) ^2 dt$	The average power is obtained by $P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) ^2 dt$
(iii)	Energy signals are non-periodic.	Power signals are periodic.
(iv)	Energy signals are time limited.	Power signals can exist over infinite time.
(v)	Examples: Pulses signals, damped signals, exponential signals etc.	Examples: sinusoidal signals, square signals etc.

chapter 2

Causality:

i. ① Causal system:

→ A system is said to be causal if the output of the system at any time t depends only on the present input and past inputs, not on future inputs.

$$\text{i.e. } y(t) = f[x(t), x(t-1), x(t-2), \dots]$$

ii. ② Anti-causal system:

→ A system is said to be anti-causal if the output of the system at any time t depends on the future inputs.

$$\text{i.e. } y(t) = f[x(t+1), x(t+2), \dots]$$

iii. ③ Non-causal system:

→ A system is said to be non-causal system if the output of the system at any time t depends on the both past, present and future inputs.

$$\text{i.e. } y(t) = f[x(t-1), x(t), x(t+1), \dots]$$

④ Significance:

→ Causality ensures that the system can be physically realized in real-time because future inputs are not available.

Stability:

→ A discrete-time system is BIBO stable if a bounded input ~~pro~~ always produces a bounded output.

Mathematically,

$$|x(n)| \leq M_x < \infty \Rightarrow |y(n)| \leq M_y < \infty$$

* Significance:

→ Stability ensures that the bounded outputs for bounded inputs.

Linearity:

→ A discrete time system is said to be linear if it satisfies

- Additivity:

$$T[x_1(n) + x_2(n)] = T x_1(n) + T x_2(n)$$

- Homogeneity:

$$T[a x(n)] = a [T x(n)] \quad \text{for any } a.$$

* Significance:

→ Linearity allows the use of superposition and convolution, simplifying analysis and design.

Time-Invariance:

→ A ~~sg~~ discrete time system is said to be time invariant if a time shift in the input ~~and~~ produces an identical time shift in the output.

Mathematically,

If $y(n) = T[x(n)]$ Then

$$T[x(n-n_0)] = y(n-n_0)$$

* Significance:

→ Time invariance ensures that the system's behavior does not change over time, making it predictable.

① Show that the impulse response of a BIBO stable LTI system is absolutely summable.

~~sol~~ let us consider a LTI system having impulse response $h(n)$ and $x(n)$ be the input applied to the system.

Then by the definition of Linear Convolution,

$$y(n) = \sum_{k=-\infty}^{\infty} x(n-k) h(k)$$

$$\Rightarrow y(n) = \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k) \quad (\because \text{By commutative property})$$

$$\therefore |y(n)| = \left| \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k) \right|$$

By swartz's inequality principle,

$$|ab| = |a| |b|$$

$$\therefore |y(n)| = \sum_{k=-\infty}^{\infty} |h(k)| \cdot |x(n-k)|$$

If the input is bounded, then $|x(n-k)| \leq M_x$

$$\therefore |y(n)| \leq M_x \sum_{k=-\infty}^{\infty} |h(k)|$$

Hence, the output will be bounded if

$$\sum_{k=-\infty}^{\infty} |h(k)| < \infty$$

$$\therefore S = \sum_{k=-\infty}^{\infty} |h(k)| < \infty \quad ; \quad S = \text{stability factor.}$$

This shows that the impulse response of a BIBO stable LTI system is absolutely summable.